

# Muhammad Ayaan Mughal

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## Summary

I work at the intersection of LLMs and real software - building the pipelines, agents, and full-stack systems that turn language models into something people can actually use. My recent work spans meeting intelligence platforms, RAG-powered automation, and agentic workflows. Finishing my CS degree at FAST-NUCES in 2026 and looking for a role where the work ships and stacks up over time.

## Education

**FAST-NUCES** Sep 2022 - Jun 2026  
B.S. Computer Science Islamabad, Pakistan  
*Relevant Coursework: Deep Learning, Generative AI, Artificial Intelligence, Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, Software Architecture, Software for Mobile Development, Cloud Computing*

## Technical Skills

|                           |                                                                                    |
|---------------------------|------------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, C/C++, Java, JavaScript, Kotlin, SQL                                       |
| <b>AI / ML</b>            | PyTorch, TensorFlow, ASR, Hugging Face Transformers, Computer Vision, scikit-learn |
| <b>LLM &amp; NLP</b>      | Prompt engineering, LLM integration, automation pipelines, RAG workflows           |
| <b>Web / Mobile</b>       | Node.js, Express, React, MongoDB, REST APIs, Android (Kotlin), Flutter, Fast API   |
| <b>DevOps &amp; Tools</b> | Git, Docker, Linux, CI/CD, n8n workflow automation, AI-assisted development        |

## Experience

**ML / AI Intern** Jun 2025 - Aug 2025  
*TechGenesys* Islamabad, Pakistan

- Built end-to-end Whisper ASR pipelines for real-time transcription, speaker segmentation, and audio preprocessing within a meeting intelligence platform. Improving transcription accuracy by 25% and reducing latency by 30
- Developed LLM-powered summarization modules with structured prompt engineering to automatically extract action items, decisions, and meeting summaries. Reducing post meeting admin time by 70%
- Developed computer vision features using OpenCV for video frame extraction; automated model evaluation and preprocessing workflows using Python, reducing manual pipeline overhead. Reducing manual attendance tracking time by 80%

## Projects

**Automated Scrum Master (Scrummate) - Final Year Project** Aug 2025 - May 2026  
*Python • n8n • LLM APIs • REST APIs*

- Multi-agent AI system that automates the full Scrum lifecycle, real-time meeting transcription with speaker diarization, LLM-extracted user stories and action items, skill-based task assignment, and a RAG interface for querying your entire project history.
- Processes 1+ hour meetings and generates sprint-ready user stories in under 30 sec; reduces post-meeting admin work by 70%. 94.3% pass rate across 35 unit tests.

**Hotel Booking Platform** 2023  
*Node.js • React • MongoDB • Express • JWT*

- Fully containerized hotel booking platform built on a microservices architecture with three independent Node.js/Express services, a shared MongoDB layer, and a unified React frontend.
- Containerised 3+ independent services with one-command local deployment and 70% automated test coverage per service.

**Autonomous Racing Agent** 2024  
*Python • TensorFlow • Deep Reinforcement Learning • TORCS*

- Hybrid manual-AI racing agent for the TORCS simulator that collects human driving data via keyboard inputs, trains an LSTM-based PyTorch model on it, then autonomously controls the vehicle in real time using supervised imitation learning.
- Trained on 10k+ driving data points over 20 epochs; agent achieves autonomous lap completion in the TORCS simulator.

**Habitify** 2025  
*Kotlin • PHP • Android • Gradle*

- Android habit tracking app with a PHP backend for persisting and syncing daily habits, streaks, and progress.